

Welcome to

Web Attacks II – Tools, Labs & Defence

Cyber Akademiet, 2026

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Slides are available as PDF, https://codeberg.org/kramse
academy-2026-day2-web-attacks-II.tex in the repo security-courses

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Goals for today



- Use assessment tools in an authorised lab: Burp, OWASP ZAP, SQLmap, Nikto
- Demonstrate web attacks against a controlled vulnerable application
- Evaluate preventive controls and communicate findings

Plan for today



- Attack proxies and scanners
- Lab: OWASP Juice Shop
- Defences: CSP, CSRF tokens, secure coding
- Reporting findings

Time schedule



- 13:15–14:45 Tools + Juice Shop – **hands-on** (booklet)
- 14:45–15:00 Break
- 15:00–16:00 Defences: CSP, CSRF tokens; reporting
- 16:00–16:15 Wrap-up

Attack proxies: Zap and Burp





OWASP recommend Zed Attack Proxy (ZAP)
https://www.owasp.org/index.php/OWASP_Zed_Attack_Proxy_Project
https://code.google.com/p/zap-extensions/

I recommend buying Burp Suite http://portswigger.net/burp/
https://pro.portswigger.net/bappstore/

Zed Attack Proxy (ZAP)



The Zed Attack Proxy (ZAP) is an easy to use integrated penetration testing tool for finding vulnerabilities in web applications.

It is designed to be used by people with a wide range of security experience and as such is ideal for developers and functional testers who are new to penetration testing.

ZAP provides automated scanners as well as a set of tools that allow you to find security vulnerabilities manually.

Source: ZAP homepage https://wiki.owasp.org/index.php/OWASP_Zed_Attack_Proxy_Project and https://www.zaproxy.org/

Burpsuite

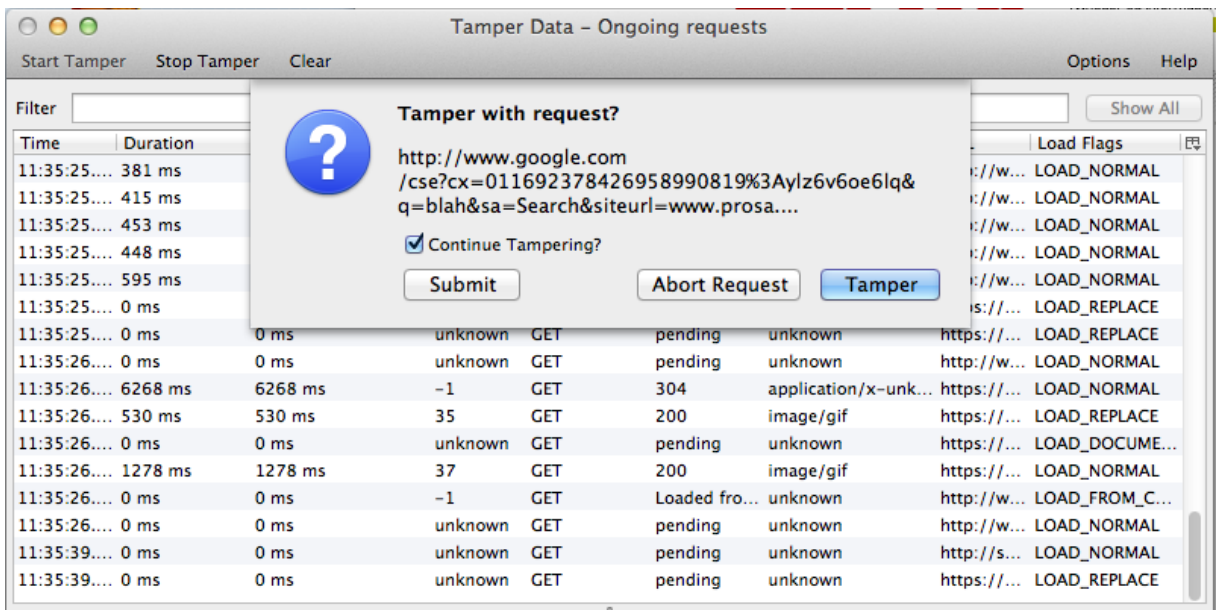


> Burp Suite is an integrated platform for performing security testing of web applications. Its various tools work seamlessly together to support the entire testing process, from initial mapping and analysis of an application's attack surface, through to finding and exploiting security vulnerabilities.
>
> Burp gives you full control, letting you combine advanced manual techniques with state-of-the-art automation, to make your work faster, more effective, and more fun.

Burp suite contains lots of functionality proxy, spider, scanner and can be extended with Java, Python programs

http://portswigger.net/burp/
https://pro.portswigger.net/bappstore/

Mini proxy: Tamper Data



Add-on for Firefox catch and modify data before it is sent to server
https://addons.mozilla.org/en-US/firefox/addon/tamper-data/

Nikto web scanner



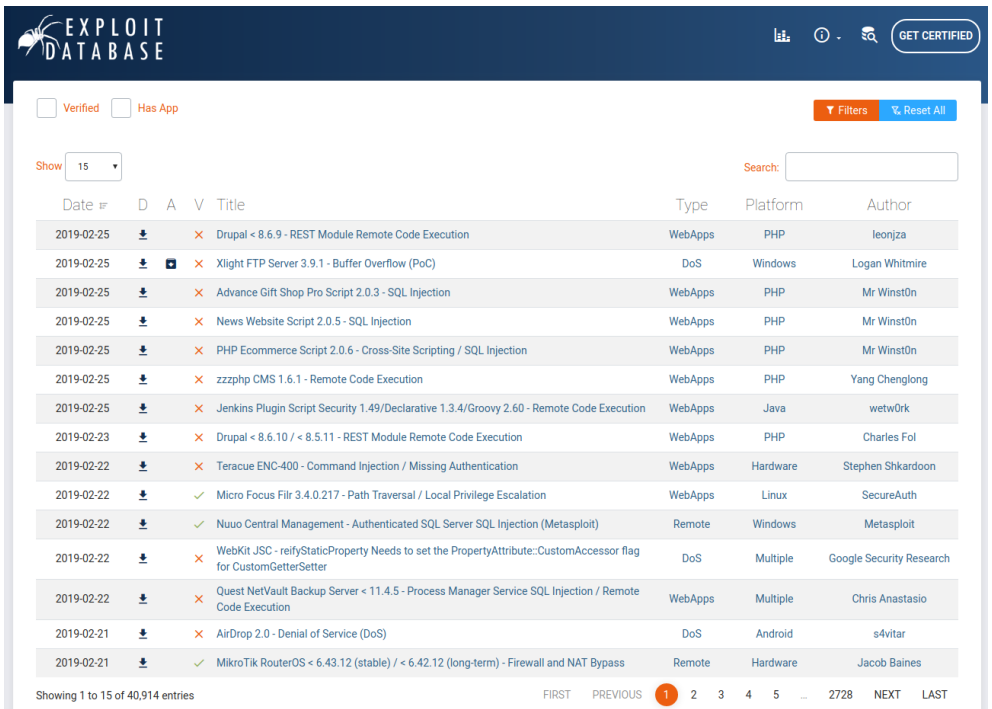
Description Nikto is an Open Source (GPL) web server scanner which performs comprehensive tests against web servers for multiple items, including over 3200 potentially dangerous files/CGIs, versions on over 625 servers, and version specific problems on over 230 servers. Scan items and plugins are frequently updated and can be automatically updated (if desired).

Quick to run, checks quite a few things. I still use and find stuff with Nikto – and you can expand it easily

```
nikto -host 127.0.0.1 -port 8080
```

Nikto web server scanner originally from http://cirt.net/nikto2

The Exploit Database



Check http://www.exploit-db.com/ and https://www.exploit-db.com/google-hacking-database and use Metasploit first!

Metasploit



> The world's most used penetration testing framework
>
> Knowledge is power, especially when it's shared. A collaboration between the open source community and Rapid7, Metasploit helps security teams do more than just verify vulnerabilities, manage security assessments, and improve security awareness; it empowers and arms defenders to always stay one step (or two) ahead of the game.

Source: https://www.metasploit.com/

- Has database support and integration with Nmap using the command `db_nmap`
- Includes various scanners: https://www.offsec.com/metasploit-unleashed/scanner-snmp-auxiliary-modules/
- Metasploit Unleashed - Free Online Ethical Hacking Course
  https://www.offsec.com/metasploit-unleashed/

OWASP Juice Shop Project



I will also use the OWASP Juice Shop Tool Project as a running example. This is an application which is modern AND designed to have security flaws.

Read more about this project at:
https://www.owasp.org/index.php/OWASP_Juice_Shop_Project
https://github.com/bkimminich/juice-shop

It is recommended to buy the Pwning OWASP Juice Shop Official companion guide to the OWASP Juice Shop from https://leanpub.com/juice-shop - suggested price USD 5.99

I run this as a docker container on Debian and it would be easy for most of you to get it running.

Brute force



We call it brute force – when testing all possibilities

```
Hydra (c) by van Hauser / THC <vh@thc.org>
Syntax: hydra [[[-l LOGIN|-L FILE] [-p PASS|-P FILE]] | [-C FILE]]
[-o FILE] [-t TASKS] [-g TASKS] [-T SERVERS] [-M FILE] [-w TIME]
[-f] [-e ns] [-s PORT] [-S] [-vV] server service [OPT]

Options:
  -S        connect via SSL
  -s PORT   if the service is on a different default port, define it here
  -l LOGIN  or -L FILE login with LOGIN name, or load several logins from FILE
  -p PASS   or -P FILE try password PASS, or load several passwords from FILE
  -e ns     additional checks, "n" for null password, "s" try login as pass
  -C FILE   colon seperated "login:pass" format, instead of -L/-P option
  -M FILE   file containing server list (parallizes attacks, see -T)
  -o FILE   write found login/password pairs to FILE instead of stdout
...
```

Cracking passwords – JtR and Hashcat



John the Ripper is a fast password cracker, currently available for many flavors of Unix (11 are officially supported, not counting different architectures), Windows, DOS, BeOS, and OpenVMS. Its primary purpose is to detect weak Unix passwords.

- Hashcat is the world's fastest CPU-based password recovery tool.
- oclHashcat-plus is a GPGPU-based multi-hash cracker using a brute-force attack (implemented as mask attack), combinator attack, dictionary attack, hybrid attack, mask attack, and rule-based attack.
- oclHashcat-lite is a GPGPU cracker that is optimized for cracking performance. Therefore, it is limited to only doing single-hash cracking using Markov attack, Brute-Force attack and Mask attack.
- John the Ripper password cracker old skool men stadig nyttig

Source:
https://hashcat.net/wiki/
http://www.openwall.com/john/

Improving the Security of your Web Applications



Start taking a critical look on your application

- Does it need to be *on the Internet* – otherwise add firewall rules!
- Does it implement HTTPS/TLS?
- What is it built using? Dependencies, frameworks, security settings
- What is the architecture – does it include a database, is the database locked down?

Content Security Policy (CSP) for XSS Prevention



The **CSP is a security configuration tool that is supported by all major browsers.** It provides settings that a developer can take advantage of to either **relax or harden security rules regarding what type of code can run inside your application.**

CSP protections come in several forms, including **what external scripts can be loaded, where they can be loaded, and what DOM APIs are allowed to execute the script**.

Let's evaluate some CSP configurations that aid in mitigating XSS risk.

- Very highly recommended, easy to check, quite easy to implement
- Tools available for you to generate even complex CSPs
- and lots of references:
  https://developers.google.com/web/fundamentals/security/csp
  https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP

Scanning for HTTP settings



Nmap can also report these settings:

```
|      X-XSS-Protection: 1;mode=block
|      Content-Security-Policy: script-src 'self' 'unsafe-inline' 'unsafe-eval'
|      X-Content-Type-Options: nosniff
```

- If you have many sites, Nmap can also report this
- Use the documentation from Mozilla Observatory for full explainations about these settings

Better: CSRF Tokens

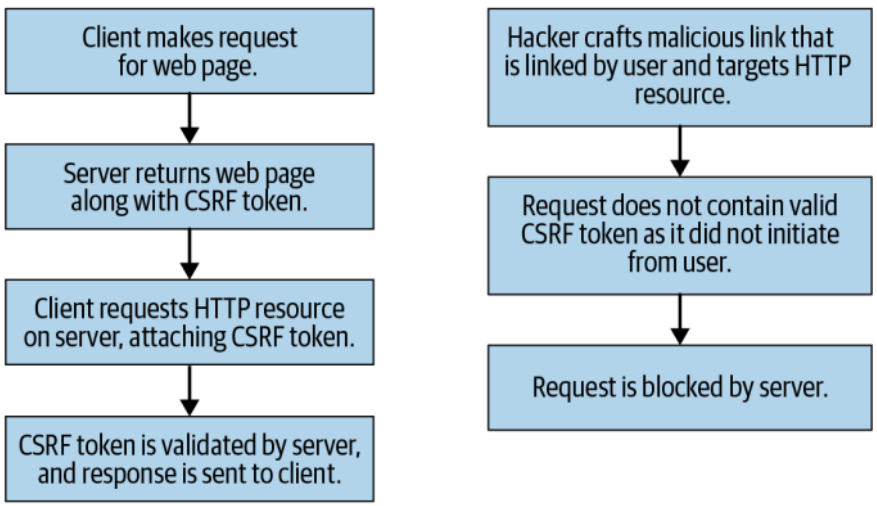


Figure 23-1. CSRF tokens, the most effective and reliable method of eliminating cross-site request forgery attacks

Source: *Web Application Security*, Andrew Hoffman, 2020, ISBN: 9781492053118

- Where do you come from, where do your browser go

Example CSRF Protection Django



Cross Site Request Forgery protection

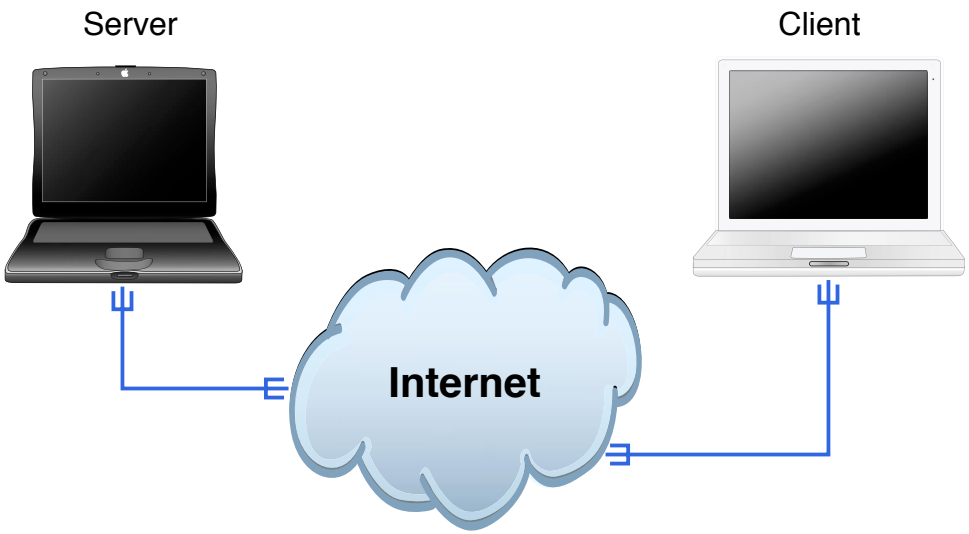
The CSRF middleware and template tag provides easy-to-use protection against Cross Site Request Forgeries. This type of attack occurs when a malicious website contains a link, a form button or some JavaScript that is intended to perform some action on your website, using the credentials of a logged-in user who visits the malicious site in their browser. A related type of attack, 'login CSRF', where an attacking site tricks a user's browser into logging into a site with someone else's credentials, is also covered.

The **first defense against CSRF attacks is to ensure that GET requests (and other 'safe' methods, as defined by RFC 7231#section-4.2.1) are side effect free**. Requests via 'unsafe' methods, such as POST, PUT, and DELETE, can then be protected by following the steps below.

Source: https://docs.djangoproject.com/en/4.1/ref/csrf/

- Many existing options can help with CSRF protection, so make sure to include this in your architecture and planning

Demo

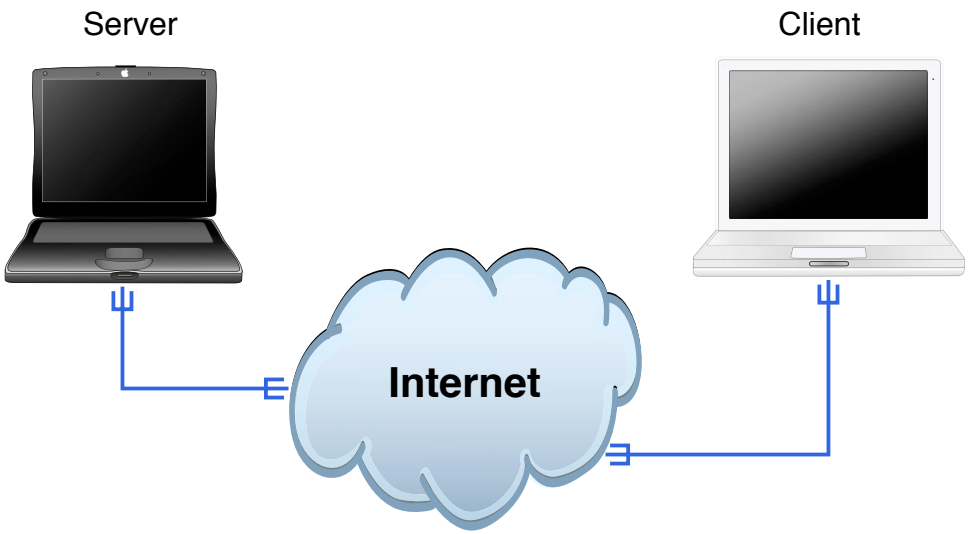


Now lets see the demo

??

which is number **??** in the exercise PDF.

Demo

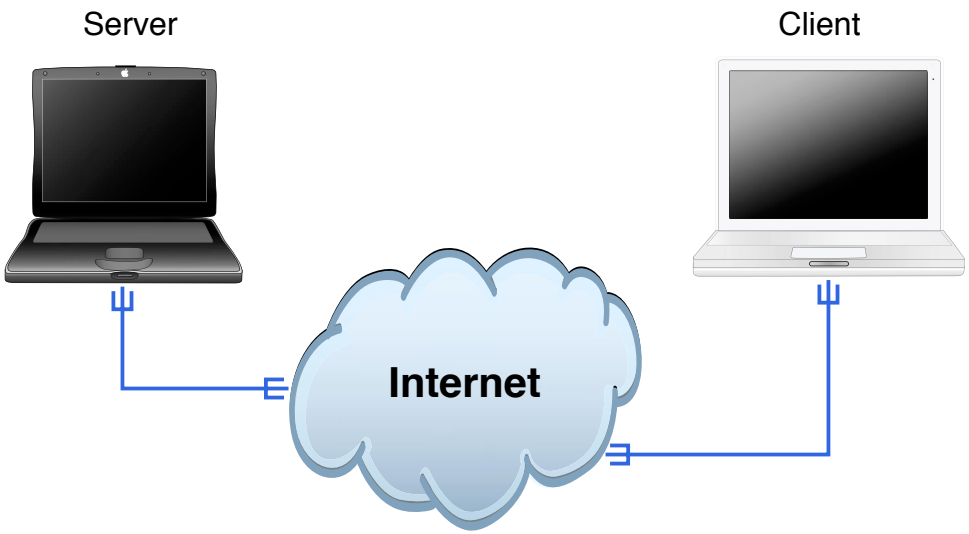


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Demo



Now lets see the demo

??

which is number **??** in the exercise PDF.

OWASP Web Security Testing Guide



The Web Security Testing Guide (WSTG) Project produces the premier cybersecurity testing resource for web application developers and security professionals.

The WSTG is a comprehensive guide to testing the security of web applications and web services. Created by the collaborative efforts of cybersecurity professionals and dedicated volunteers, the WSTG provides a framework of best practices used by penetration testers and organizations all over the world.

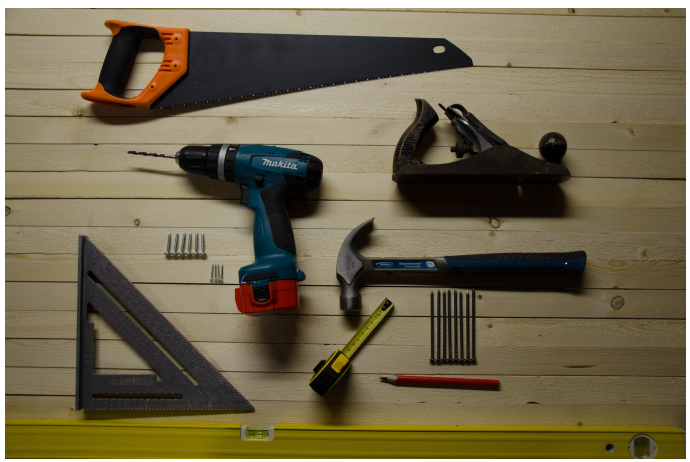
- Project from OWASP:
  https://owasp.org/www-project-web-security-testing-guide/
- Use the Tab *Release Versions* to download version 4.2 in PDF
- Also available as a checklist `OWASPv4_Checklist.xlsx`

Reporting findings – technical & management



- **FIXME:** document evidence, impact, prioritised remediation (obj 11)

Conclusion Pentest web attacks



- You really should try testing, and investigate your existing devices all of them
- This is just one small part of your security posture
- Harden servers, networks, configurations
- Also block outgoing connections – shellshock, log4shell, ... next CVE coming

Questions?



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You are always welcome to send me questions later via email

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